

Development of an ingredient containing apple peel, as a source of polyphenols and dietary fiber.

Apple peel is a waste product from dried apple manufacture. The content of phenolic compounds, dietary fiber, and mineral are higher in apple peel, compared to other edible parts of this fruits. The objective of this study was to develop an ingredient from Granny Smith apple peel, using a pilot scale double drum-dryer, as drying technology. The control of all steps to maximize the retention of phenolic compounds and dietary fiber was considered. Operational conditions, such as drying temperature and time were determined, as well as important preprocessing steps like grinding and PPO inhibition. In addition, the physical-chemical characteristics, mineral and sugar content, and technological functional properties such as water retention capacity, solubility index, and dispersability among others, were analyzed. A simple, economical, and suitable pilot scale process, to produce a powder ingredient from apple peel by-product, was obtained. The drying process includes the application of ascorbic acid at 0.5% in the fresh apple peel slurry, drum-dryer operational conditions were 110 degrees C, 0.15 rpm and 0.2 mm drum clearance. The ingredient developed could be considered as a source of phenolic compounds (38.6 mg gallic acid equivalent/g dry base) and dietary fiber (39.7% dry base) in the formulation of foods. Practical Application: A method to develop an ingredient from Granny Smith apple peel using a pilot scale double drum-dryer as drying technology was developed. The method is simple, economical, feasible, and suitable and maximizes the retention of phenolic compounds and dietary fiber present in the raw matter. The ingredient could be used in the formulation of foods.